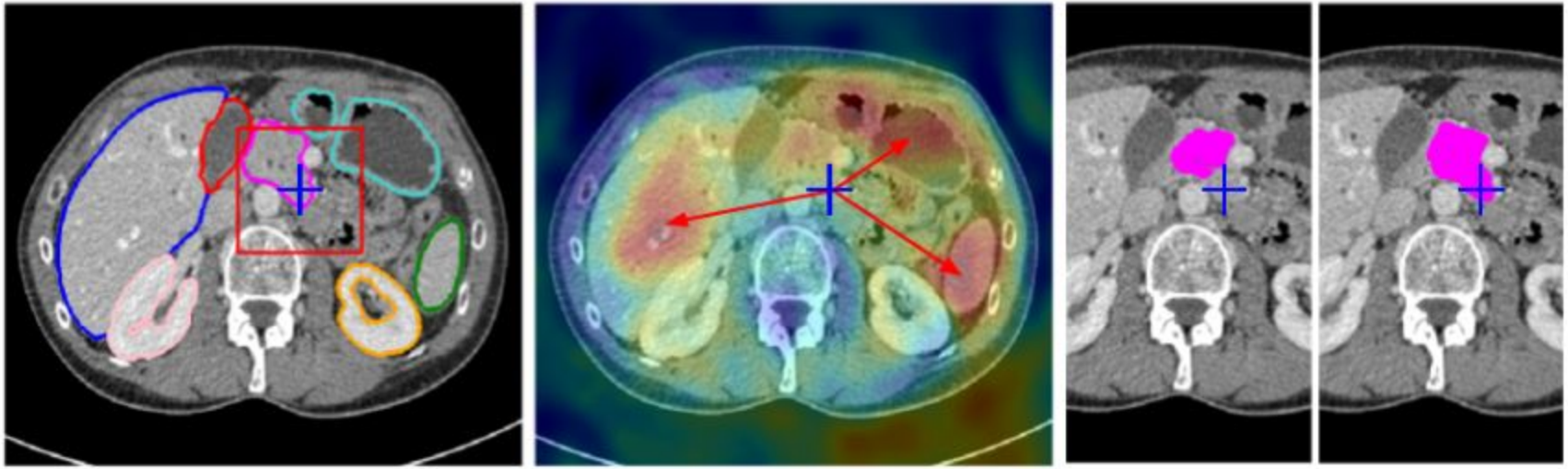
Lecture 5

Self and Cross Attention

U-Net Transformer: Self and Cross Attention for Medical Image Segmentation

https://arxiv.org/pdf/2103.06104



a) Ground Truth b) Attention map c) U-Net d) U-Transformer

Fig. 1. Global context is crucial for complex organ segmentation but cannot be captured by vanilla U-Nets with a limited receptive field, *i.e.* blue cross region in a) with failed segmentation in c). The proposed U-Transformer network represents full image context by means of attention maps b), which leverage long-range interactions with other anatomical structures to properly segment the complex pancreas region in d).

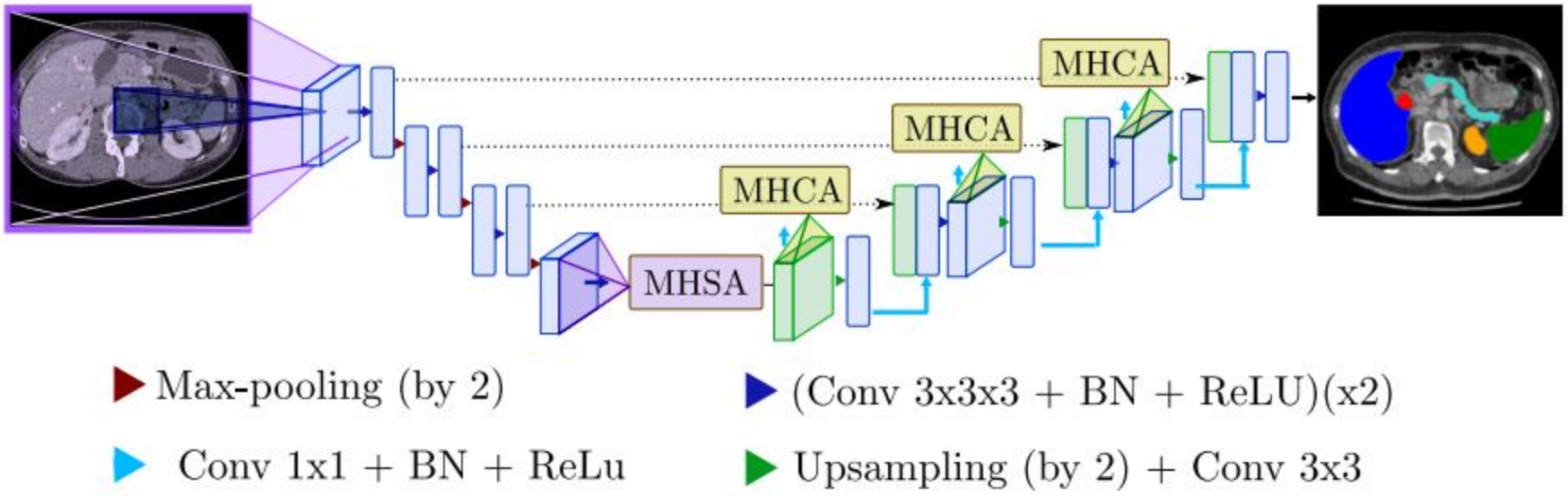


Fig. 2. U-Transformer augments U-Nets with transformers to model long-range contextual interactions. The Multi-Head Self-Attention (MHSA) module at the end of the U-Net encoder gives access to a receptive field containing the whole image (shown in purple), in contrast to the limited U-Net receptive field (shown in blue). Multi-Head Cross-Attention (MHCA) modules are dedicated to combine the semantic richness in high level feature maps with the high resolution ones coming from the skip connections.

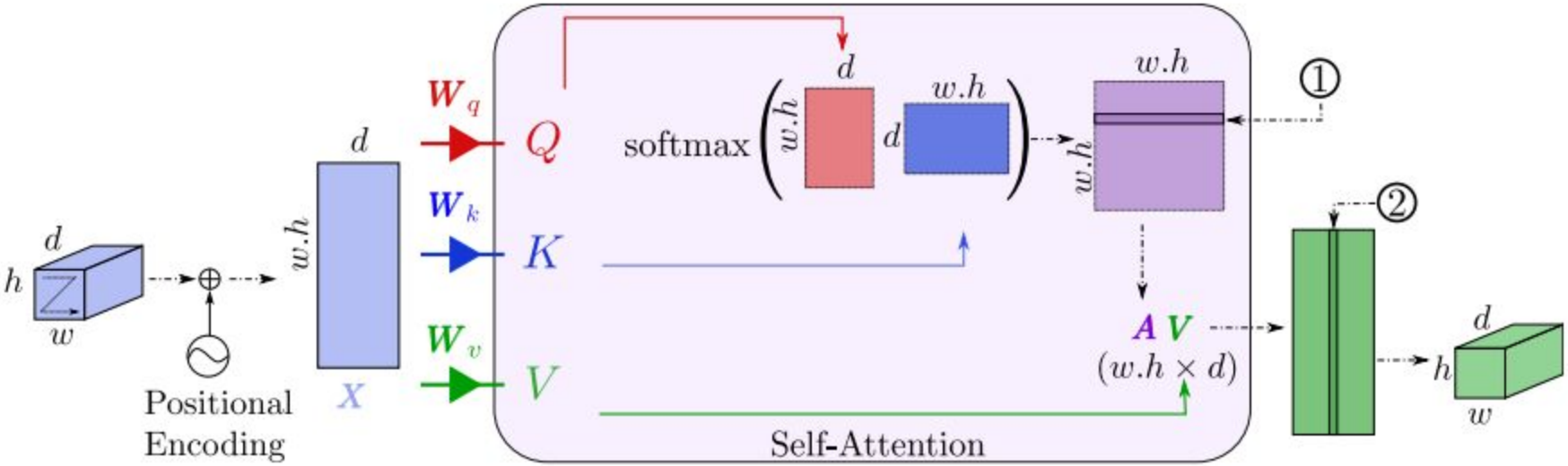


Fig. 3. MHSA module: the input tensor is embedded into a matrix of queries $\boldsymbol{Q}$, keys $\boldsymbol{K}$ and values $\boldsymbol{V}$. The attention matrix $\boldsymbol{A}$ in purple is computed based on $\boldsymbol{Q}$ and $\boldsymbol{K}$. (1) A line of $\boldsymbol{A}$ corresponds to the attention given to all the elements in $\boldsymbol{K}$ with respect to one element in $\boldsymbol{Q}$. (2) A column of the value $\boldsymbol{V}$ corresponds to a feature map weighted by the attention in $\boldsymbol{A}$.

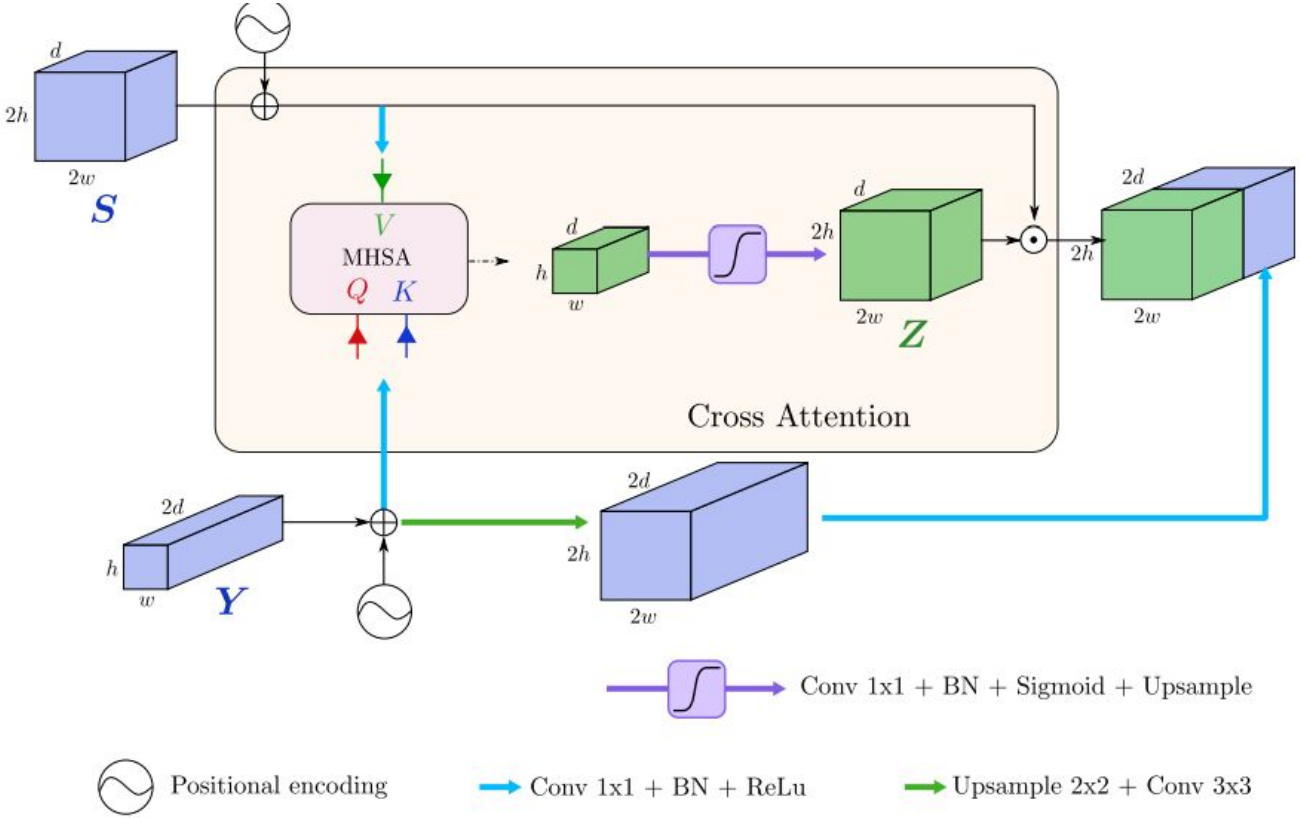


Fig. 4. MHCA module: the value of the attention function corresponds to the skip connection $\boldsymbol{S}$ weighted by the attention given to the high level feature map $\boldsymbol{Y}$. This output is transformed into a filter $\boldsymbol{Z}$ and applied to the skip connection.

Table 1. Results for each method in Dice similarity coefficient (DSC, %)

Dataset	U-Net [11]	Attn U-Net [13]	MHSA	MHCA	U-Transformer
TCIA	76.13 (± 0.94)	76,82 (± 1.26)	77.71 (± 1.31)	77.84 (± 2.59)	**78.50** (± 1.92)
IMO	86.78 (± 1.72)	86.45 (± 1.69)	87.29 (± 1.34)	87.38 (± 1.53)	**88.08** (± 1.37)

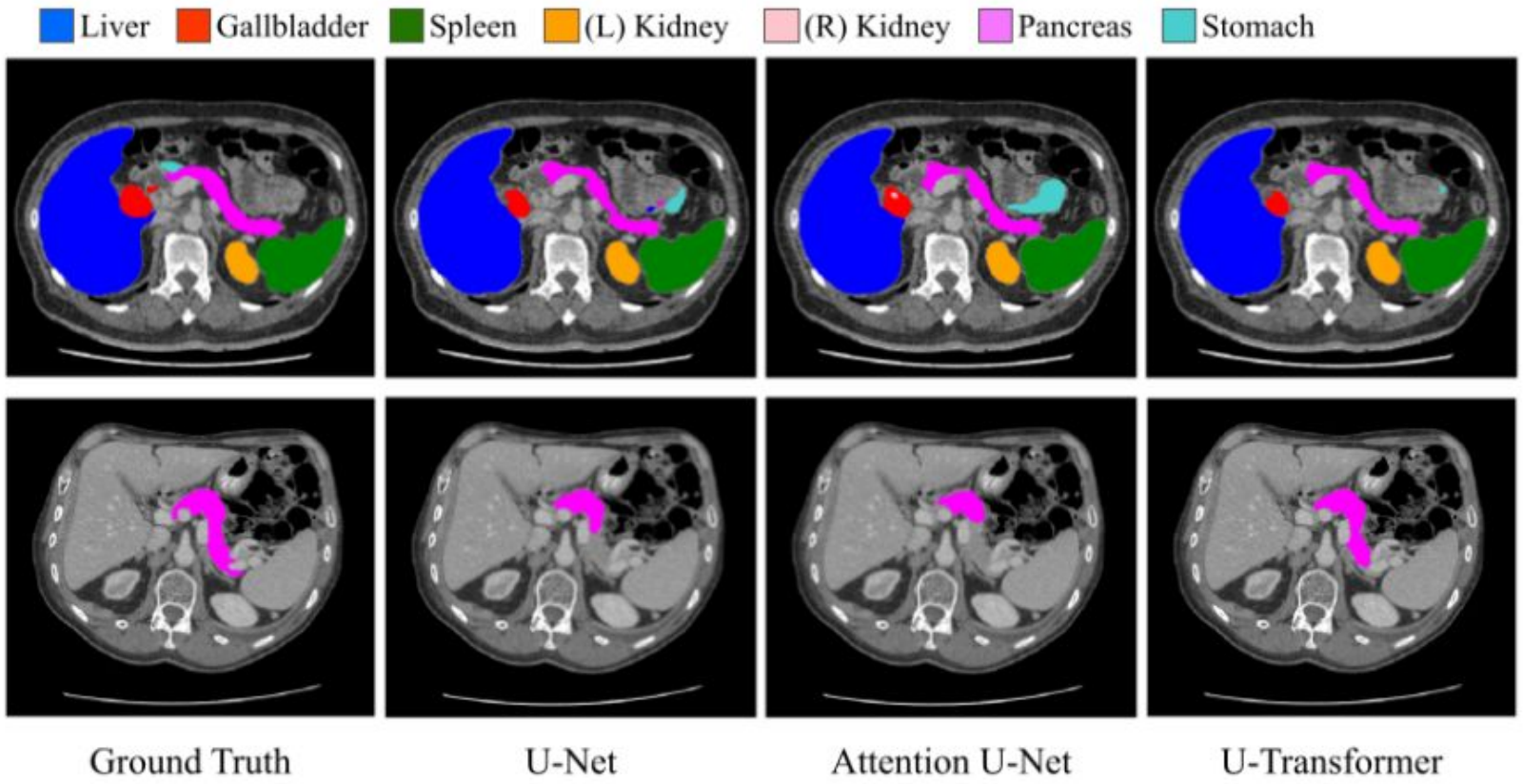


Fig. 5. Segmentation results for U-Net [11], Attention U-Net [13] and U-Transformer on the multi-organ IMO dataset (first row) and on TCIA pancreas (second row).

Table 2. Results on IMO in Dice similarity coefficient (DSC, %) detailed per organ.

Organ	U-Net [11]	Attn U-Net [13]	MHSA	MHCA	U-Transformer
Pancreas	69.71 (± 3.74)	68.65 (± 2.95)	71.64 (± 3.01)	71.87 (± 2.97)	**73.10** (± 2.91)
Gallbladder	76.98 (± 6.60)	76.14 (± 6.98)	76.48 (± 6.12)	77.36 (± 6.22)	**78.32** (± 6.12)
Stomach	83.51 (± 4.49)	82.73 (± 4.62)	84.83 (± 3.79)	84.42 (± 4.35)	**85.73** (± 3.99)
Kidney(R)	92.36 (± 0.45)	92.88 (± 1.79)	92.91 (± 1.84)	92.98 (± 1.70)	**93.32** (± 1.74)
Kidney(L)	93.06 (± 1.68)	92.89 (± 0.64)	92.95 (± 1.30)	92.82 (± 1.06)	**93.31** (± 1.08)
Spleen	95.43 (± 1.76)	95.46 (± 1.95)	95.43 (± 2.16)	95.41 (± 2.21)	**95.74** (± 2.07)
Liver	96.40 (± 0.72)	96.41 (± 0.52)	96.82 (± 0.34)	96.79 (± 0.29)	**97.03** (± 0.31)

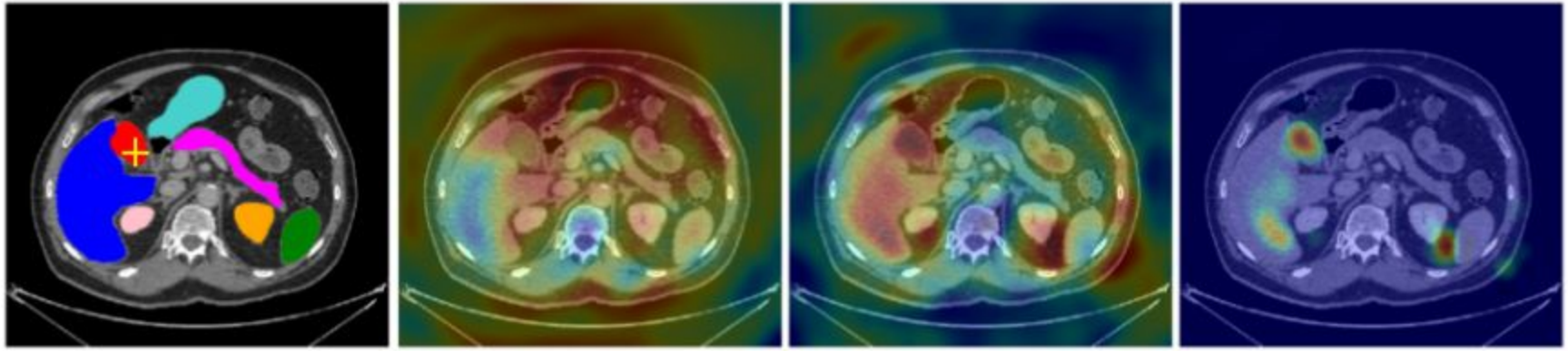


Fig. 6. Cross-attention maps for the yellow-crossed pixel (left image).

Table 3. Ablation study on the positional encoding and multi-level on one fold of TCIA and IMO.

			MHSA		MHCA		
	U-Net	Attn U-Net	wo PE	w PE	1 lvl wo PE	1 lvl w PE	multi-lvl w PE
TCIA	76.35	77.23	78.17	**78.90**	77.18	78.88	**80.65**
IMO	88.18	87.52	88.16	**88.76**	87.96	88.52	**89.13**

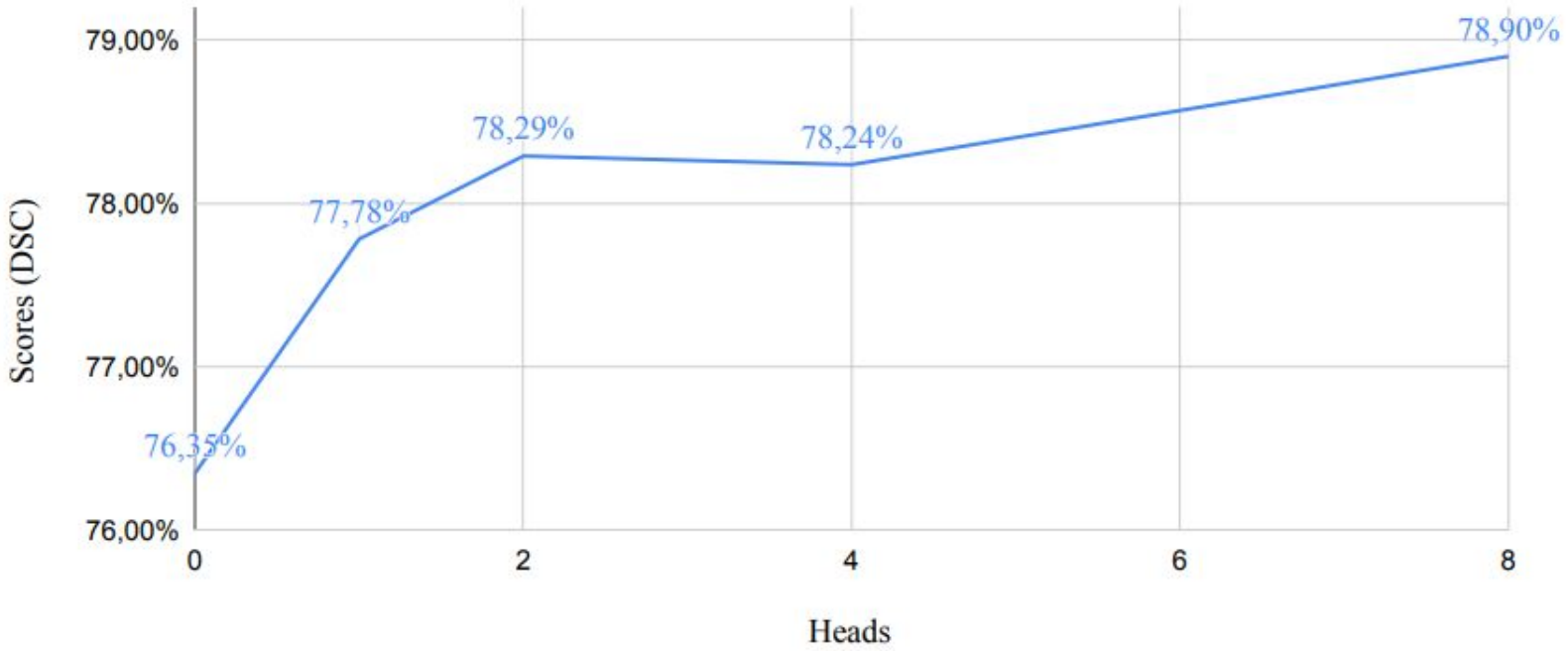


Fig. 7. Evolution of the Dice Score on TCIA (fold 1) when the number of heads varies between 0 and 8 in MHSA.

Table 4. Hausdorff Distances (HD) for the different models

Dataset	U-Net	Attn U-Net	U-Transformer
TCIA	13.61 ($\pm$ 2.01)	12.48 ($\pm$ 1.36)	**12.34** ($\pm$ 1.51)
IMO	12.06 ($\pm$ 1.65)	12.13 ($\pm$ 1.58)	**12.00** ($\pm$ 1.32)